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Risk-shifting

This question is designed to show you how to apply insights from option pricing to corporate finance. It also underscores the how convexity interacts with volatility.

I. THE ALE IDENTITY AND EQUITY AS AN OPTION

Consider a firm with risky assets given by a random variable V and debt with face value F. The firm is protected by limited liability: it repays if its assets are worth more than the face value of its debt—if $V \geq F$—and otherwise it defaults. In this case, the debt holders get the assets.

1. Call the payoff of debt D_T. What is D_T if $V \geq F$? What if $V < F$? Sketch a graph of D_T. Write an expression for D_T.

2. Consider the assets V as the underlying of a derivatives position. What is the name of the position with the payoff D_T?

3. Call the payoff of equity E_T. Use the balance sheet (ALE) identity to write down an expression for E_T in terms of V and D_T. Use the graphs of V and D_T to sketch E_T.

4. Show (graphically or algebraically) that $E_T = \max\{0, V - F\}$.

5. Considering the assets V as the underlying of a derivatives position, what is the name of the position with the payoff E_T.

6. Is the payoff E_T convex, concave, or neither?

II. RISK-SHIFTING

1. Suppose that the risk-free rate is zero and that the firm has \$10 of debt and \$10 of cash, i.e. $V = 10$ and $F = 10$. What is the expected value of the firm's assets? What is the variance? What is the payoff D_T of debt? What is the payoff E_T of equity? What is the value of equity today (before T)?

2. Suppose that the firm gets an unexpected investment. It costs \$10 and has a random payoff. It pays off \$20 with probability $1/4$ and zero with probability $3/4$. What is the expected value of the firm's assets if it undertakes the investment? What is their variance?

3. Suppose that the firm undertakes the investment. What is $\mathbb{E}[D_T]$, the expected payoff of debt? What is $\mathbb{E}[E_T]$, the expected payoff of equity?

4. Suppose, as in reality, that equity holders have the right to choose what the firm does. Do equity holders choose to do the investment or not? Comment on why. Is this good for debt holders? If not, what should debt holders do to protect themselves?